

Coordination and Performance in Global Software Service Delivery: The Vendor's Perspective

Anandasivam Gopal, J. Alberto Espinosa, Sanjay Gosain, and David P. Darcy

Abstract—Past research on software development outsourcing has explored in some detail the client's perspective and its attendant performance issues. However, relatively few studies have explicitly addressed performance drivers viewed from the vendor side. In this paper, we fill this gap by examining how coordination within the project team and between the client and the vendor organizations influence two dimensions of software project performance—software quality and development speed—using data from 83 software projects from nine Indian software firms. Our results show that both client (external) coordination and vendor team (internal) coordination positively influenced software quality, but not development speed. We also found that client communication barriers moderated the impact of coordination with the client on quality. While we did not find that vendor-team coordination affected development speed, team size had a negative interaction effect with vendor-team coordination on development speed. Interestingly, temporal boundaries had a detrimental effect on software quality, but a positive effect on development speed. Finally, we found that development speed increased with the number of person days devoted to the project up to a certain point (i.e., a first-order effect), but speed declined with additional person days (i.e., a second-order effect). Our findings contribute to literature and practice by offering nuanced insights into performance in outsourcing projects from the vendor perspective and the role of coordination, both within the vendor team and with the client.

Index Terms—Coordination, development speed, offshoring, outsourcing, quality, regression analysis, software development.

I. INTRODUCTION

"In the sometimes panicked desire to save money—especially with the powerful lure of 'half-price' workers in places like India, China, and the Philippines—good execution flies out the window. And that's where the problems flock in. Outsourcing is not for the faint-hearted or the ill-prepared. It just doesn't 'happen'." [1]

OFFSHORE outsourcing has taken on an increased pace in recent years driven by the impulse to take advantage of

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A. Gopal is with the Robert H. Smith School of Business, University of Maryland, College Park, MD 20742 USA (e-mail: agopal@rsmith.umd.edu).

J. A. Espinosa is with the Kogod School of Business, American University, Washington, DC 20016-8200 USA (e-mail: alberto@american.edu).

S. Gosain is with the Capital Group Companies, Inc., Irvine, CA 92618 USA (e-mail: gosain@gmail.com).

D. P. Darcy is with the Irish Management Institute, Dublin 16, Ireland (e-mail: drdavep@gmail.com).

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cost efficiencies, but it has also resulted in damaging conditions captured by the quote above. The recent economic downturn has further spurred the quest for efficiencies through restructuring of IT activities involving some type of offshore software development. Given the scale of the phenomenon, it is not surprising that offshore outsourcing continues to attract considerable theoretical and practical attention. At the same time, current economic pressures have imposed on both vendors and clients the need to carefully examine factors that impact the performance of outsourcing engagements more carefully. In this paper, we address one such determinant of performance in the *software services outsourcing* context—coordination. While coordination, outsourcing and global team performance have been studied individually in the literature, this paper makes a unique contribution by bringing these three perspectives together.

In any market-based activity, coordination is critical for contracting parties to manage their relationship effectively. From the perspective of an offshore outsourcing client of software services, a natural concern is how to effectively manage the arrangement with the vendor such that business and technical requirements are met; indeed, some of the insightful case work on offshore engagements has studied this aspect in some detail [2], [3]. However, from the vendor's perspective, the key concern is how to develop quality software in a timely and efficient manner such that the vendor's payoff is also protected. This vendor perspective is different from the client's [4], and it is understudied in the offshore outsourcing context. Investigating these issues from the vendor's perspective may lead to some nuanced results about what drives performance in an offshore outsourcing arrangement.

In this specific context, there are distinct differences in the manner in which a vendor provides services, compared to the in-house software delivery context, including: 1) the development team not only needs to manage its activities internally within the team but also manage work activities externally with the client [5]; 2) the vendor's coordination needs are likely to be different with the client than within the vendor development team—e.g., coordination with clients may focus on ensuring accurate exchange of requirements information, whereas coordination within teams may focus on expedient delivery of the software product; 3) part of the team is separated by geographic distance at the client's site; and 4) there are global boundaries (e.g., geographic, cultural, organizational) that need to be bridged to get the work done [6], [7].

In a typical offshore outsourcing project, the initial high-level requirements are drawn out by a small vendor team located at the client site to gather requirements, which the client approves [8]. These general requirements are then elaborated into fully

developed requirements specifications offshore at the vendor's site where the majority of the system is then developed in close consultation with designated client liaisons [9]. A small part of the development team often stays onshore to liaise with the client and provide further information and support. Once the final system is completed, this small team becomes responsible for installation and acceptance testing after which the warranty period begins. This project arrangement poses several challenges in terms of coordination [5] and subsequent implications for project success. While offshore outsourcing of software services can be cost-effective for clients and profitable for vendors, the coordination needs can be daunting for large projects. Moreover, coordination needs to take place across geographical, temporal, and cultural boundaries, while bridging communication barriers associated with interorganizational differences. Hence, we explore the following research questions, which have high practical and theoretical relevance.

- 1) How do vendor team (internal) coordination and client (external) coordination influence project performance in offshore software services delivery?
- 2) How do client communication barriers and global boundaries moderate the relationship between coordination and project performance?

Our research contributes to the literature in several ways. First, we add to the growing body of knowledge on globally outsourced software development. Second, we model and empirically examine the *differential* role of vendor team and client coordination on project performance. Third, we examine how global boundaries and communication barriers enhance or offset the effects of coordination on performance. Fourth, we explicitly consider the impact of coordination on two relevant performance indicators from the vendor's perspective—project duration and software quality. While a client perspective may consider other performance criteria pertinent (such as customer satisfaction or total cost), vendors are driven more by how effective and efficient the service delivery process is [10].

The rest of the paper is organized as follows. Section II reviews relevant literature streams and Section III describes theory and research hypotheses, while the research methodology, data collection, and analysis approach is discussed in Section IV. The results from our analysis, contributions to managers, and limitations of our work are included in Section V. Section VI concludes the paper.

II. LITERATURE REVIEW

A. Coordination in Global Outsourcing

The issue of coordination has received significant attention in studies of global software development [11]–[13]. However, most of the extant research has focused on either in-house development, or on outsourcing projects from the client's perspective and the literature is relatively sparse when it comes to the performance implications of *coordination* from the vendor's perspective. Little empirical work exists that has addressed these issues in any detail to understand the vendor's tradeoffs with respect to coordination along different performance dimensions on the project. With the exception of some insightful case studies that

highlight the role of coordination [2], [5], [14], [15], there is a significant gap in the literature, which our work addresses.

Coordination is necessary to manage dependencies among task activities carried out by various actors to integrate the work [16]. In the context of our study, there is a natural division of tasks that fall into three groups: the client, the vendor team on site, and the vendor team offshore. Therefore, we focus how these groups coordinate—i.e., internally within the vendor team and externally with the client. Prior research has characterized this in the form of two types of key challenges—i.e., the “organizational challenge” or expertise coordination within the vendor team—and the “relational challenge” or knowledge exchange between the vendor and the client [5]. Reports in the practitioner press also point to the importance of both types of coordination in managing outsourced projects [17]–[19] discuss several dimensions of coordination that are possible, such as formal/informal [20] and impersonal/interpersonal [21], with various coordination mechanisms available within each form of coordination. In this paper, we focus on planned, formal coordination for two reasons. First, research has shown that informal coordination is prevalent in software development [20], but the ability for collaborators to interact informally tends to be hampered with global boundaries and organizational differences that are observed in the case of outsourcing [12]. Second, formal coordination is critical in outsourcing work because it is enshrined in many of the process models that are followed by software service delivery vendors, such as the capability maturity model (CMM) [22], thereby making it most relevant in our study. Our focus here is on the emergent coordination mechanisms, which manifest themselves in the actual conventions that arise during the project [23].

We also differentiate our model by taking into account the unique set of coordination needs faced by the offshore software vendor team because its members need to bridge multiple global boundaries to get the job done. These boundaries create discontinuities and possible “fault lines” [24] in the work environment that increase the coordination overhead [6], [7]. In order to coordinate internally within the team, members will need to bridge spatial and often temporal boundaries. In contrast, to coordinate externally with clients, team members need to bridge cultural boundaries and overcome communication barriers arising from interorganizational differences. These differences between internal and external boundaries can cloud our vision of how coordination influences project performance, which our study attempts to clarify.

B. Coordination and Performance in Software Development

1) *Coordination*: Software development is an inherently complex task with many interdependencies. In order to achieve high performance in a collaborative software task, these dependencies need to be well managed, which is by definition the *raison d'être* for coordination [16]. The classic organizational literature has long established that coordination is very important in organizations [21], [25]. The need for coordination in software development is not new [15], [20], [26], [27]. Prior research suggests that coordination in software projects should

lead to better performance, all else being equal [20], [28], [29], particularly in global contexts [11], [30]. However, coordination needs vary among groups carrying out different task activities with different types of dependencies in a software project. For example, a recent study [31] found that project managers are more sensitive to process and temporal dependencies (such as adherence to processes and synchronization of schedules), whereas developers are more sensitive to technical dependencies (interoperability of software parts). Thus, it is natural to expect that coordination outcomes in different groups will differ depending on the specific coordination needs of these groups. As discussed above, there are three main groups in the global outsourcing context with different intergroup dependencies that need to be managed through formal coordination mechanisms: clients, vendor team on-site, and vendor team offshore.

This positioning highlights two relevant aspects discussed in this paper. First, it requires the separation of client coordination (external) from vendor coordination (internal), since they address different sets of dependencies. Second, it suggests explicitly linking formal coordination to specific outcomes, since certain outcome dimensions are likely affected by different dependencies [19]. Therefore, we posit that client (external) coordination is driven by client requirements and thus more focused on improving product quality, whereas vendor (internal) coordination is driven by productivity goals and thus more focused on improving efficiency or development speed.

2) *Performance*: Performance in offshore software delivery borrows from conceptualizations in software development research and is typically measured along two dimensions—*process* and *product performance*. Process performance measures the efficiency of the development process [32]—i.e., whether the project is completed within budget and on schedule [33]. Product performance captures the extent to which the system delivered meets requirements [32]. While it is possible to study each dimension individually, it is insightful to use both dimensions of performance concurrently since it is often easy to cut duration by underperforming on quality [34]. Therefore, we focus our attention on two performance indicators in this paper—quality (representing product performance) and project development speed (representing process performance).

a) *Software Quality*: Quality in software services outsourcing is typically associated with the extent to which the delivered software services adhere to customer requirements or to certain characteristics that are important to the client [35]. Extant research also shows that quality is one of the central parameters of interest in the outsourcing market [10]. Unlike in-house software development, outsourced software projects tend to have very strong service-oriented characteristics [8]. In such contexts, the definition of quality becomes multifaceted, which is reflected in the many characterizations of quality seen in the literature. For instance, Wallace *et al.* [36] used five criteria for software quality: reliability, maintainability, meeting requirements, and response time of the application developed. Others advocate more objective criteria like defects per line of code [35] or bug fixes per modification request [13]. Because of this service-oriented nature of software outsourcing, we conceptualize quality in terms of desirable software properties sought

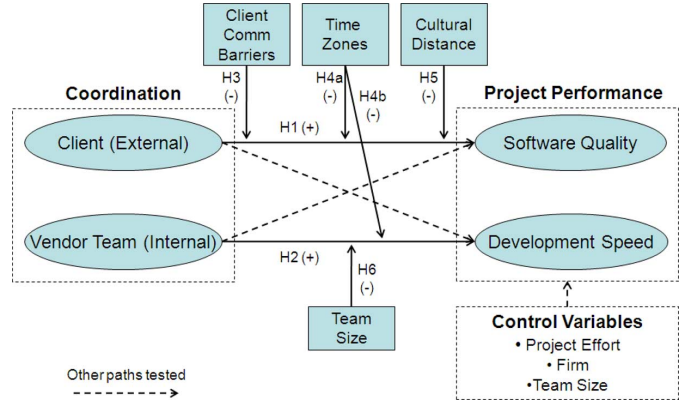


Fig. 1. Research model.

by clients such as reliability, flexibility, and ease of use. These properties, rather than defect rates, are more closely associated with client requirements in services and are consistent with conceptualizations seen in the literature [37], [38].

b) *Development Speed*: We conceptualize development speed as a measure of the efficiency with which a team is able to complete the execution of the software project, all else being equal. This is represented by the calendar time elapsed between the time the project was signed on and its completion date, which is consistent with prior studies [13], [39]. However, since development speed is influenced by the amount of resources assigned to a project, we control for the total project effort (i.e., person-days expended on the project) to make our results comparable across projects with different platforms, environments, programming languages [33], and resource levels. Development speed is also a key performance indicator for IT service vendors [22] because it is associated with the vendor's project costs and profitability [9], particularly in competitive environments where time-to-market is critical.

III. THEORY AND HYPOTHESES

The conceptual model presented in this paper is shown in Fig. 1. We explicitly hypothesize differing effects of client and vendor coordination in the offshore context. Specifically, we argue that client coordination will provide a significant benefit on quality of software services delivered, whereas the primary benefit of vendor coordination will be on improving speed on the project. Similarly, we postulate differing effects of client communication barriers and global boundaries as moderators of the relationship between coordination and performance. We provide the reasoning for the specific hypotheses below.

A. Client (External) Coordination

Because of the service orientation in outsourced software development, we argue that client coordination is more likely to focus on managing technical and other product dependencies needed to meet client requirements. Therefore, we anticipate that the first-order benefit for the vendor from higher client coordination will be better information exchange, which will primarily lead to improved software quality. Curtis *et al.* [40]

found that “communication and coordination breakdowns” were a major problem affecting software productivity and quality in large systems development, and that breakdowns often occurred at organizational boundaries, thereby suggesting that coordination between organizations was important for project success. Client coordination is necessary to manage project dependencies that pertain to exchange and sharing of information between the client and the vendor team. In typical outsourcing software projects, these include things like: vendor liaisons to facilitate communication with the client; project documents and specifications to ensure that requirements are communicated precisely and unambiguously; project schedules; and procedures to track key issues. Service provider vendors possess the technical knowledge needed to develop the software, but not the firm- and context-specific knowledge needed for effective requirements analysis [26]. Continuous coordination is required as the project reaches maturity [14], [41], and technical details (e.g., incremental releases of code) and other project management information (e.g., project reports, status reviews) need to be provided by the vendor to the client on a regular basis [9].

Software development teams face significant coordination challenges when working with client organizations [42]. Difficulties in knowing who to contact about what and how to initiate contact and communicate effectively may lead to undetected conflicts among assumptions and incorrect interpretation of communications [42]. Unless remedied through rework, these differences will manifest in the form of defects or limited functionality. In addition, open issues not resolved early in the project may escalate into quality problems later on [43]. These various problems will surface if dependencies among the task activities are not appropriately managed through formal coordination protocols between the client and vendor team. Indeed, Vlaar *et al.* [44] studied how these protocols can lead to congruence in action between vendors and clients, leading to greater sense-making on the project and a more tacit understanding of the various issues surrounding the project.

Research on new product development teams has also shown that a lack of coordination in the presence of dependencies leads to mistakes [45]. On the other hand, coordination can not only help organizations manage dependencies, but also help the client and the vendor work more independently on their respective task activities and help develop agreement upon common goals with sufficiently clear task subgoals for everyone [29], thus improving the quality of information exchanged. Hence, we propose the following hypothesis.

Hypothesis 1: Greater exercise of client (external) coordination will lead to higher software quality.

B. Vendor Team (Internal) Coordination

Vendors serve multiple clients, and therefore, have a motivation to be productive. Thus, we argue that vendor-team coordination is more focused on managing temporal and process dependencies *within* the team to improve process efficiency, resulting in better schedule control and increased development speed. Mintzberg [46] proposed that coordination could be facilitated by standardization of work processes, outputs, skills,

and knowledge to enable better management of dependencies across activities and allowing actors in an organizational system to work on collective tasks. The importance of internal team coordination has been empirically studied with product development teams (e.g., [47]). Vendor-team coordination requires the presence of mechanisms for software project planning and execution, as well as the presence of appropriate and planned feedback to managers about deviations from plan, along with a commitment to the plan and readiness to make prompt corrections to the plan when needed. It also requires strict adherence to formal processes adopted for the project. Collectively, these coordination mechanisms should help the vendor team meet project schedules on time.

Kraut and Streeter [48] found that lack of coordination could result in software projects taking longer to complete. One cause often identified with such delays is “feature churn” [27] whereby developers spend unnecessary time and effort developing functionality later considered unnecessary. Research on distributed software-development projects also confirms that cross-site coordination issues lead to loss of development speed in handling modification requests [42]. Project requirements need to be precisely communicated to the on-site vendor team and these have to be effectively relayed over to the development team offshore. In turn, vendors need to communicate the status of development activities and the evolving vision of the software system back to the client thorough the same channel. This information needs to be quickly and efficiently exchanged between the on-site and off-site members of the vendor team, which underscores the importance of internal coordination for the timely completion of the project. Vendor-team coordination will also ensure that work issues and conflicts that arise over the course of a project are addressed in a timely and effective manner without prolonging the project.

Prior research with software teams suggests that team coordination helps improve performance [20] and reduce the inherent risk [28] in the development process. Increased vendor-team coordination will ensure that members can work concurrently on their respective tasks, thus reducing time spent on interdependency issues [29] with less work overlap and redundant work. It is also interesting to note that vendor-team coordination is one of the key process areas in the CMM and has been shown to significantly enhance the management of internal dependencies between various tasks, leading to enhanced efficiency [9]. Hence, we posit the following hypothesis.

Hypothesis 2: Greater exercise of vendor (internal) team coordination will lead to higher development speed.

In our theorizing, we have assumed that the two forms of coordination are distinct but not uncorrelated. It is possible that both vendor team and client coordination have a common root, either in firm-level processes or in environmental uncertainty. Therefore, it is conceivable that client (external) coordination could influence development speed and vendor team (internal) coordination could influence software quality. However, increased attention to client coordination may divert productive vendor resources away from the development task, thereby hindering development speed [49]. Similarly, the impact of vendor-team

coordination on software quality may not be substantial above and beyond what client coordination accomplishes. Consequently, we test but do not offer any predictions on these effects. We simply argue that client and vendor-team coordination are fundamentally aimed at resolving different sets of dependencies observed in the offshore software context.

C. Client Communication Barriers

We argued above that client (external) coordination has a positive effect on software quality. However this effect is likely to vary depending on the quality of communication between the client and the vendor team. One distinguishing characteristic of offshore outsourcing arrangements is the presence of interorganizational differences between the vendor team and the client, which can hinder communication [22]. Such interorganizational differences are likely to become salient [50] making coordination between the vendor team and the client difficult to achieve without some level of shared understanding between them [51]. In the offshore context, existing challenges to collaborative work are exacerbated, among other things, by differences in internal culture and norms of social behavior [17], [26], [52] attitudes toward authority between the client and vendor organizations [26]. There may also be differences in system development methodologies and practices between the client and vendor organizations in the offshore market [53], creating additional hurdles to cross to manage work processes effectively and avoid miscommunication to develop quality software that meets client requirements. Also, transferring domain knowledge and requirements information to the offshore vendor is expected to be more challenging in the presence of interorganizational differences [41], [54]. In sum, we argue that client communication barriers resulting from interorganizational differences will attenuate the dissemination and processing of coordination information needed to ensure product quality. Lack of knowledge about the client's business and methodologies will adversely affect the ability of the development team to share project requirements information unambiguously with the client. Additionally, managerial differences existing between clients and vendor teams will continue to reduce the efficacy of planning and feedback mechanisms. Therefore, we propose the following hypothesis.

Hypothesis 3: Client communication barriers due to interorganizational differences will negatively moderate the relationship between client coordination and software quality.

D. Global Boundaries

Global boundaries make coordination more challenging since traditional interpersonal forms of coordination, such as encounters in hallways and water coolers, are unavailable [20]. Difficulties in knowing who to contact about what, initiating contact, and communicating effectively or safely with client teams or managers, can lead to unrecognized conflicts among assumptions and incorrect interpretation of communications [42]. One important characteristic that distinguishes vendor team (internal) coordination from client (external) coordination in our context is the presence of global boundaries in the collaborative task. As described before, it is the responsibility of the ven-

dor team to coordinate locally with colocated team members while concurrently coordinating with a client liaison located onshore [9]. The offshore model introduces two types of global boundaries that add coordination overhead: temporal boundaries imposed by differences in time zones between the vendor and client locations; and cultural boundaries between the vendor and client countries. Together, these boundaries align creating a "fault line" [24] that is often difficult to bridge. We discuss these next.

1) *Temporal Boundaries:* Maznevski and Chudoba [55] showed that work in effective global teams center around incidents of intense interactions, punctuated by lower volumes of interactions through leaner media, with team activities organized around repeating patterns of such incidents. However, this ability to interact in a synchronized and spontaneous manner is diminished with temporal separation, which is an important characteristic of globally outsourced projects. In a simulation study, Espinosa and Carmel [56] modeled the various costs that time differences between team members or organizational units impose on coordination and showed that there are significant costs associated with communication and clarification processes, often leading to delays and rework. Sarker and Sahay [52] also studied distributed software teams and found that there are significant problems that emerge from differences in time zones. One strategy suggested to address time zone issues is to institute norms of communication and make work-processes visible through planned and more formal coordination. van Fenema and Qureshi [57] also pointed to the fact that working hours in India overlap minimally with those of the U.S. client organizations, virtually eliminating interactive exchanges that may be required to solve complex problems. Such conditions create coordination problems [30], which often lead to disjointed knowledge spheres, impairing the ability of client and vendor teams to work toward achieving quality outcomes. Therefore, we have the following hypothesis.

Hypothesis 4a: Temporal boundaries will negatively moderate the relationship between client coordination and software quality.

An interesting aspect of offshore projects is the geographic configuration of vendor teams that provide software services to the client. As discussed earlier, while the bulk of the development team remains "offshore," a small team remains at the client site in order to carry out requirements analysis, flesh out systems specifications, liaise with the client, and help resolve open issues during the project [8], [22]. Thus, for a significant part of the project, the vendor's development team is also affected by the presence of temporal boundaries within the team. The resulting time-zone differences delay work completion and reduce the effectiveness of communication [58]. Time-zone differences also reduce the window of opportunity for synchronous interaction within the team. Clarifications and open issue resolution are typically done a day later and often lead to disruptive working times for team members in different time zone [56], particularly when there is little or no overlap in work hours [30]. van Fenema and Qureshi [57] reported on these challenges: "Time zone may be different, so there is always a delay in getting answers. With China we do not have much of a problem, but between US

and Singapore we do. Sometimes when they (colleagues from China) ask me a question I may not be able to answer them, I have to forward it to the US. By the time they give me a reply, it's already night there, it's day here. He may not understand my question properly. So he may not give the right answer. He may just come up please elaborate more,' or 'give me please one day since I have to go back.'" Instead of being able to utilize real-time phone conversations and chats that would solve problems in a short timeframe, members on the vendor team have to shift toward a more asynchronous communication patterns that lengthen "problem-solving" cycles and lower development speed. Therefore, we have the following hypothesis.

Hypothesis 4b: Temporal boundaries will negatively moderate relationship between vendor-team coordination and development speed.

2) *Cultural Boundaries*: Practitioner articles (cf., [59]) suggest that misunderstandings and problems stemming from cultural differences are endemic to global outsourcing projects—as many as 51% of the CIOs surveyed reported that the greatest "offshore" outsourcing challenge is overcoming cultural differences. Recent work on understanding the relevant decision criteria for clients looking to outsource their IT work has also identified the presence of cultural differences as a significant source of uncertainty [10]. Hofstede's seminal work on national cultures has shown systematic variations among people belonging to different nations [60]. Even simple cultural differences [61] between partners in economic activity can influence business outcomes in international contexts like joint ventures [62]. These cultural differences have important implications for work coordination between the vendor team and the client's personnel. In contrast to Western hierarchical relationships that tend to be governed by contractual agreements and an ideology of essential equality, Indian hierarchical relationships are oriented toward firmly internalized expectations in both superiors and subordinates for reciprocity and mutual obligations in a more closely connected emotional relationship [63]. Similarly, Asian subordinates may require more guidance than American managers tend to provide [64]. Research has shown that cultural differences affect work patterns such as forms of partnership and coordination mechanisms [65]. Experiments have shown that teams from low uncertainty avoidance cultures may be more risk-seeking and susceptible to escalation of commitment behavior [66]. In offshore projects, this dispersion in attitudes and values can lead to conflict between vendor and clients.

Anecdotal evidence points to the inefficiencies imposed by cultural differences noted above. For instance, Overby [67] noted: "One reason for that is the American workers' comfort level with speaking up and offering suggestions. A good American programmer will push back and say 'what you're asking for doesn't make sense, you idiot.' Indian programmers have been known to say, 'this doesn't make sense, but this is the way the client wants it.'" Such cultural differences will impose additional coordination requirements because of the need to repeat and clarify things that are well-understood in their native cultures, which if unresolved can lead to misunderstandings and lower software quality. Thus, cultural differences between the vendor team and the client will lead to some misalignment in

expectations about the project and less mutual understanding, offsetting the positive effect of client coordination on software quality. Therefore, we have the following hypothesis.

Hypothesis 5: Cultural boundaries will negatively moderate the relationship between client coordination and software quality.

3) *Team Size*: Considerable research in software development teams has identified the role played by team size in influencing project outcomes [13], [68] particularly with geographically dispersed teams, which has been found to affect performance [13]. However, there is less direct evidence of the impact of team size on coordination with respect to process efficiency. While adding more members to a software team brings additional individual productive resources, it also makes it more difficult for all team members to interact and work together to integrate their individual work into a single working product. The number of possible dyads and that need to interact and share information in a team to manage their respective dependencies increases exponentially with team size (i.e., $n(n-2)/2$, [68]).

Highly interdependent tasks like software development require substantial information sharing and team communication [12]. Each dependency link between developers will demand time and effort that can make it more difficult to coordinate effectively. For example, as the team increases in size, the amount of familiarity that an additional team member needs to have or acquire about existing members will be higher [39]. It will also be more difficult to know which members know what in larger teams, which research has found to be important for performance [69], thus increasing the need to become more familiar with team members. Therefore, while we argue that vendor-team coordination is critical in ensuring development speed, we also recognize that the size of the team can affect this relationship. Therefore, we have the following hypothesis.

Hypothesis 6: Team size will negatively moderate the relationship between vendor-team coordination and development speed.

IV. RESEARCH METHODOLOGY

A. Data Collection

We collected structured survey data for this study from Indian firms because of the leading role that Indian firms are playing in the global software services arena over the last 10 years. In order to homogenize the sample, given the wide variety of economic activities being outsourced [70], we used several criteria to focus our sampling strategy. First, the choice of a single country helped control for country differences such as culture and infrastructure. Second, we selected companies with either a CMM Level 3 or ISO 9001 certification from a database of almost 600 software firms maintained by NASSCOM (an industry-level software services trade association). Third, we only focused on software-development outsourcing where the activity being outsourced was turnkey software projects. Applying these controls provided us with a set of 267 companies. We selected randomly 45 firms as potential data collection sites and contacted their senior management to request their participation. Once they agreed to participate, we requested a list of representative software projects that had been completed in the

TABLE I
SAMPLE DESCRIPTION ($n = 83$)

Project Characteristic	Sample Descriptives
Software Domain	Application 54 (65.1%); System 14 (16.9%)
Client Industry	Biotech/Healthcare 9 (10.8%); Finance 7 (8.4%); Information Technology 29 (34.9%); Automobile, transport and Logistics 9 (10.8%); Manufacturing 7 (8.4%); Retail 4 (4.8%)
Project Type	New Development 43 (51.8%); Maintenance 16 (19.3%); Reengineering 15 (18.1%)
Platform	Mainframe 14 (16.9%); Unix variants 26 (31.3%); Windows 25 (30.1%)
Client Country	Japan 7 (8.43%); UK 3 (3.61%); USA 56 (67.47%) ; Other (Australia, France, Germany, Singapore, Sweden and Switzerland) 17 (32.5%)

Note: Remaining projects could not be clearly classified into one of these categories; they belong to additional categories.

previous six months. We used this short time frame to reduce effects of recall bias and to minimize issues associated with attrition of managers, which is a significant problem in India. All projects chosen were outsourced and carried out in full by the vendor firm.

A total of 23 firms agreed to participate in our study yielding a response rate of 51%. Organizations that declined to participate indicated time commitments and other institutional priorities as their reason. We obtained complete data on a sample of 119 projects from these firms, comprising development, reengineering, and maintenance software services projects. Some of the firms in our sample only provided a single project, which would increase the chances of sampling bias and possibly confound the firm-level and project-level causal effects—i.e., we would be drawing both firm-level and project-level inferences from one observation, thereby increasing the level of bias in the sample. To eliminate these problems and to ensure that we had broad representation from each firm in the final sample, we only retained projects from firms that provided data on five or more projects and controlled for firm effects. The final sample for this study consisted of 83 projects from nine firms. Table I provides a descriptive summary of the projects included in the sample.

We collected data using a survey questionnaire administered to project managers in charge of the projects, which we created using scales from prior research to capture data on coordination and boundaries, and on project-level metrics about work effort and duration. The majority of the scales used was reflective in nature and were used as such in prior research. We first pretested the questionnaire with a group of quality auditors certified by the Software Engineering Institute (SEI), whose feedback was incorporated into the questionnaire. We then created a web-based version of the questionnaire and conducted a pilot survey with 15 software organizations in India (not included in final sample). The data in the final survey instrument were collected using seven-point Likert scales. The operationalization of the variables is described below. The specific items used are shown in the Appendix.

1) *Software Quality*: Measures the extent to which the final software product conforms to client requirements. We used a taxonomy of software quality attributes [71] to identify four important quality attributes required in the offshore development market, similar to quality measures used in prior re-

search [36], [38]. These four items (Cronbach's $\alpha = 0.71$) are also consistent with definitions of software quality used in prior software development research [28], [29].

2) *Development Speed*: It is measured as the project duration or cycle time in calendar days. A $Q-Q$ plot for this variable revealed that it was skewed to the left. We investigated several Box-Cox transformations and a natural logarithm transformation yielded the best approximation to a normal distribution. Also, we reversed the sign of this variable (by multiplying it by -1) to make it easier to interpret results—i.e., higher values represent higher development speed.

3) *Client Coordination*: This is measured using four questionnaire items adapted from work on coordination in software-development projects [20], [43]. These items captured the extent to which coordination mechanisms (by plan, by feedback, and through integrative roles) were followed between the client and vendor teams (Cronbach's $\alpha = 0.75$).

4) *Vendor-Team Coordination*: It captures the presence of formal project management and process-improvement procedures within the project team and is adapted from past work [43], [72]. It is measured using five items pertaining to existence of coordination procedures (by plan and by feedback) within the development team (Cronbach's $\alpha = 0.78$).

5) *Client Communication Barriers*: These are measured using three questionnaire items (Cronbach's $\alpha = 0.70$) created based on a discussion of offshore coordination boundaries in Krishna *et al.* [53], which identified communication problems due to interorganizational differences. Additionally, we used descriptions of communication problems experienced in offshore development by practitioners in wording the questionnaire items [73], [74].

6) *Cultural Boundaries*: It is based on data reported by Hofstede on different cultural dimensions for each country (http://www.geert-hofstede.com/hofstede_dimensions.php). These dimensions are power distance, individualism, masculinity, uncertainty avoidance, and long-term orientation. We constructed a distance metric following prior work [75] based on average squared difference in scores for each dimension of culture between the client country and India.

7) *Temporal Boundaries*: It is measured as the absolute time-zone difference [30] between the client country location and India.

TABLE II
SUMMARY STATISTICS AND PAIRWISE CORRELATIONS ($n = 83$)

	Mean	S.D.	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
a. Client Coordination	6.090	0.844								
b. Vendor Team Coordination	5.575	0.866	0.523**							
c. Client Communication Barriers	2.281	1.207	-0.133	-0.127						
d. Team Size	2.473	0.736	-0.038	-0.034	0.222*					
e. Temporal Boundaries	8.582	3.086	-0.126	-0.088	-0.387**	0.044				
f. Cultural Boundaries	876.750	154.848	0.039	0.115	-0.138	-0.331**	-0.128			
g. Project Effort	7.027	1.313	0.070	-0.047	0.036	0.268*	-0.105	-0.283*		
h. Software Quality	5.914	0.574	0.470**	0.507**	-0.326**	-0.349**	-0.142	0.305**	-0.291*	
i. Development Speed	-3.622	3.811	-0.147	-0.231	0.061	0.163	-0.029	-0.173	0.635**	-0.300*

* Correlation significant at 0.05 (two tailed), **correlation significant at 0.01 (two tailed); summary statistics reported are for uncentered variables.

TABLE III
EXPLORATORY FACTOR ANALYSIS ($n = 83$)

Questionnaire Item	Component		
	Factor 1	Factor 2	Factor 3
VENDCOORD1	0.727	0.226	0.033
VENDCOORD2	0.724	0.063	-0.150
VENDCOORD 3	0.750	0.289	-0.027
VENDCOORD 4	0.622	0.380	0.025
VENDCOORD5	0.809	0.113	-0.071
CLIENTCOORD1	0.131	0.758	-0.310
CLIENTCOORD2	0.180	0.710	0.034
CLIENTCOORD3	0.375	0.769	-0.020
CLIENTCOORD4	0.174	0.632	0.068
BARRIER1	-0.073	0.070	0.799
BARRIER2	-0.082	0.076	0.832
BARRIER3	0.023	-0.308	0.648

Extraction Method: principal component analysis. Rotation method: Varimax with Kaiser normalization. Cumulative variance extracted = 60.4%

8) *Team Size*: It captures the core project team size on the project. The distribution of team size in our sample was positively skewed and therefore, we also used a log transformation.

9) *Project Effort*: It was measured using total person-days expended on the project, as a proxy for project size. We used a log transformation for this variable.

B. Data Analysis

The correlation matrix and descriptive statistics for our sample are presented in Table II. We first ran factor analysis on the 12 questionnaire items used to operationalize client coordination mechanisms, vendor-team coordination mechanisms, and global boundaries. The results, which are reported in Table III, show that the items load well on the factor underlying the hypothesized construct with factor loadings of over 0.6 and cross-loadings lower than 0.4, indicating discriminant validity [76]. In order to further establish discriminant and convergent validity, we use Venkatraman's [77] methodology—specifically, we tested for convergent validity and unidimensionality and found that a single latent variable underlay a set of measures. We tested for this by estimating three individual confirmatory factor models for the independent variables—client coordination, ven-

dor coordination, and global boundaries. The results, shown in Table IV, indicate strong statistical properties with goodness of fit indices in the acceptable regions [78] for all three constructs, indicating strong unidimensionality and convergent validity.

Following Venkatraman [77], we also tested for discriminant validity by evaluating factor models with pairs of constructs in each case. We evaluated the two-factor model in each case with a free covariance parameter between the two latent factors first (unconstrained model). We then fixed the correlation between the two latent factors at unity and evaluated the (constrained) model. The difference in fit between the two models shows whether the factors show adequate discriminant validity. In our context, the confirmatory factor analysis results of the three pair-wise analyses are shown in Table V. In each case, the Chi-square change between constrained and unconstrained models is significant at the 0.01 level, indicating good discriminant validity. In addition, we report the internal consistency reliability indicator [77] for our measures. This reliability indicator captures the ratio of the latent construct variance to the construct plus error variance. Venkatraman [77] has indicated that composite reliability of 0.50 and above are acceptable for constructs (see Table IV). All our constructs meet this threshold with composite reliability of 0.70 or higher.

As recommended by Podsakoff and Organ [79], we elicited the dependent variable questionnaire items before the items for independent variables. We tested for common method bias with Harmon's single factor test [79] using factor analysis on all items for dependent, independent and control variables. We obtained multiple factors with eigenvalues greater than 1, with the largest factor accounting for less than 25% of the variance. Podsakoff and Organ [79] also suggested a more conservative test, in which the first factor is partialled out and the hypothesized results are examined. We performed this test using the factor scores for the first factor as additional predictors in the regression. Including this factor did not lead to any significant changes in our model results, suggesting that common method bias is not a major concern. Nevertheless, to be conservative, we further tested for common method bias by administering a short survey recollecting data for software quality using the same quality scales, 18

TABLE IV
UNIDIMENSIONALITY AND CONVERGENT VALIDITY TEST RESULTS ($N = 83$)

Factor	Composite Reliability	No. of indicators	χ^2	d.f.	$\chi^2/\text{d.f.}$	GFI	AGFI	RMR	CFI
Vendor Team Coordination	0.821	5	6.91	5	1.382	0.97	0.90	0.02	0.98
Client Coordination	0.760	4	0.91	2	0.45	0.98	0.97	0.02	0.99
Client Communication Barriers ^a	0.700	3	20.31	19	1.06	0.94	0.90	0.08	0.98
Client Communication Barriers ^b	0.700	8	18.49	13	1.42	0.94	0.89	0.08	0.96

Note: a, b – The communication barriers model is saturated since there are three indicators. Therefore, fit indices for this variable were produced from two-factor models run with Vendor Team Coordination (a) and Client Coordination (b).

TABLE V
DISCRIMINANT VALIDITY OF COORDINATION AND BOUNDARY VARIABLES ($n = 83$)

Construct pair	ML estimate ϕ	t-value	χ^2 Statistic		
			Constrained model (df)	Unconstrained model(df)	Difference
Vendor Team - Client Coordination	0.41	3.44*	30.63 (27)	22.79 (26)	7.84**
Vendor Team Coord. – Client Comm Barriers	-0.08	-0.64	37.54 (20)	19.96 (19)	17.58**
Client Coord – Client Comm Barriers	-0.07	-0.66	33.27 (14)	17.26 (13)	16.01**

Note: * - $p < 0.05$, ** - $p < 0.01$.

months after the original data collection. We received 24 responses of which 20 were complete (24% of original sample). The new quality measures using our quality items were highly correlated with the old quality data for these projects ($r = 0.73$, $p < 0.001$), attesting to the validity of our quality measure. We averaged the item scores pertaining to each construct to form a composite measure for the construct and used in regression analysis for hypotheses testing. We estimated two equations, one each for software quality and development speed. Prior research suggests that the relationship between project effort and development speed is nonlinear [80], therefore, we introduced a quadratic term for project effort in the development speed equation, shown as follows:

$$\begin{aligned} \text{Software quality} = & f(\text{client coordination, vendor} \\ & \text{– team coordination, client communication barriers,} \\ & \text{cultural boundaries, temporal boundaries, team size,} \\ & \text{project effort, client coordination} \times \text{client} \\ & \text{communication barriers, client coordination} \times \\ & \text{cultural boundaries, client coordination} \times \text{temporal} \\ & \text{boundaries}) + \varepsilon_1 \end{aligned} \quad (1)$$

$$\begin{aligned} \text{Development speed} = & f(\text{client coordination, vendor} \\ & \text{– team coordination, client communication barriers,} \\ & \text{cultural boundaries, temporal boundaries, team size,} \\ & \text{project effort, project effort}^2, \text{ vendor – team coordination} \\ & \times \text{temporal boundaries, vendor – team coordination} \times \\ & \text{team size}) + \varepsilon_2. \end{aligned} \quad (2)$$

The interaction terms capture the moderating effects of global boundaries. In order to minimize the effects of multicollinearity from the interaction terms, we centered all interacting vari-

ables [81]. Centering the dependent variable and the interacting variables eliminates problems with scale invariance with multiplicative terms and provides more meaningful interpretation of main effects [82]. We estimated (1) and (2) using ordinary least squares (OLS). We examined the data for OLS assumptions—i.e., normality of residuals, outliers [81], and independence and found no violations of the assumptions. In order to capture the firm-specific fixed effects, we included eight binary control variables for the firms in both equations. We also clustered the variances around the firm in order to capture any nonindependence of the error terms across projects from the same firm. The clustered standard errors are robust to correlations across observations from the same firm and provide more conservative standard errors. The results of the OLS regression models are shown in Table VI. Both models are statistically significant. The R^2 for the Quality and development speed equations are 0.515 and 0.405, respectively, indicating high explanatory power. The variance inflation factors were under 2 and the condition indices were under 10, indicating no problems with multicollinearity. In conducting the moderated regression analyses, we adhered to recommendations provided by Carte and Russell [83]. Judgment of moderation was based on whether the change in R^2 was significant.

V. RESULTS AND DISCUSSION

A. Results

Table VI shows the results of hierarchical regressions for software quality and development speed. Because none of the organizational binary variables were significant and did not add significant predictive power to the regression models, we removed them from both models. In support of Hypothesis 1,

TABLE VI
RESULTS OF HIERARCHICAL REGRESSION ANALYSIS, $n = 83$

Variables	DV = Software Quality		DV = Development Speed	
	Model 1: Direct Effects	Model 2: Full Model	Model 1: Direct Effects	Model 2: Full Model
Coordination				
Client Coordination	0.173 ** (0.065)	0.237 *** (0.068)	-0.048 (0.080)	-0.039 (0.078)
Vendor Team Coordination	0.190 ** (0.063)	0.169 ** (0.062)	-0.013 (0.078)	-0.008 (0.078)
Barriers and Boundaries				
Client Communication Barriers	-0.130 ** (0.044)	-0.141 *** (0.043)	0.014 (0.055)	0.025 (0.055)
Temporal Boundaries	-0.035 * (0.017)	-0.034 * (0.017)	0.033 (0.021)	0.049 * (0.022)
Cultural Boundaries	0.048 (0.052)	0.027 (0.062)	0.019 (0.064)	0.002 (0.064)
Other Variables				
Team Size	-0.140 * (0.070)	-0.155* (0.068)	-0.094 (0.091)	0.157 □ (0.095)
Project Effort	-0.097 * (0.041)	-0.087 * (0.041)	0.899 * (0.351)	1.304 ** (0.398)
Project Effort ²			-0.092 *** (0.027)	-0.121 *** (0.030)
Interactions				
Client Coordination X Client Communication Barriers		-0.112 ** (0.041)		
Client Coordination X Temporal Boundaries		-0.012 (0.022)		
Client Coordination X Cultural Boundaries		0.001 (0.001)		
Vendor Team Coordination X Temporal Boundaries				-0.025 0.027
Vendor Team Coordination X Team Size				-0.177 □ (0.091)
Adjusted R ²	0.471	0.515	0.384	0.405
ΔR ²	-	0.044	-	0.021
F-value	11.420 ***	9.713 ***	7.397 ***	6.587 ***

□ $p < .10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; Regression coefficients are unstandardized (standard errors in parentheses); Controls for organizational indicators were non-significant in both quality and development speed models.

we found that client coordination influenced software quality. Surprisingly, vendor-team coordination did not influence development speed, as predicted in Hypothesis 2, but it did have a positive effect on software quality. This is intriguing because it suggests that coordination in the outsourcing context is focused on sharing requirements information that will lead to improved software quality and that coordination does not have development speed benefits. Intuition and prior research suggests that coordination improves productivity, but managing dependencies is costly and the time spent on this is time diverted from development activities. So we speculate that vendor team developers focus their attention on what matters the most to their clients—i.e., delivering high quality products that meet their requirements. This does not mean that speed is not important, but that perhaps there are more effective ways to reduce software development duration. According to our results, one such way revolves around sizing the project optimally. Interestingly, we found a significant effect for the quadratic modeling of project effort, suggesting that sizing the project at an optimal number of person days may yield more speed benefits than anything else in outsourcing. Because the first-order effect is positive and the second effect is negative, more person-day resources added to the project improve speed up to a certain optimal point be-

yond which the negative effect of the quadratic effort variable reduces development speed, perhaps due to the high coordination overhead typically associated with very large, complex projects [68].

Another factor influencing speed is the presence of time zones. Our results show that temporal boundaries have a significant negative effect on quality, but a positive significant effect on development speed. This result is consistent with prior experimental studies in the laboratory [84], which have shown that developers can work faster with some amounts of time separation, but that quality is negatively affected. This would explain why coordination is focused on managing dependencies that affect quality, so as to offset the detrimental effects of temporal separation, which is not the case with development speed. Although prior research in outsourcing has discussed the impact of time zones, this study is among the first to quantify a direct effect of these impacts on quality.

The interaction effects are also interesting. In the quality model the predictive power of the regression model increased significantly ($p < 0.001$) when the interaction terms were added to the model. As shown in Fig. 2, and in support of Hypothesis 3, there was a significant negative interaction between client communication barriers (due to interorganizational differences)

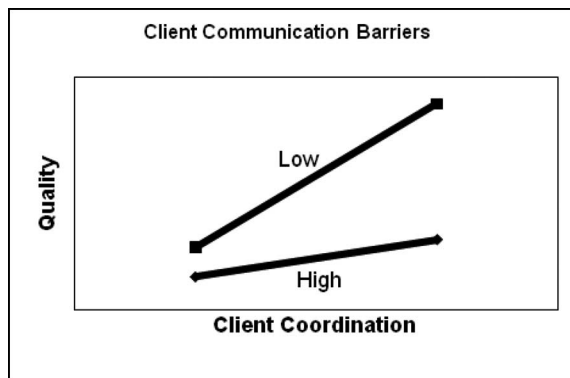


Fig. 2. Interaction between client coordination and communication barriers.

and coordination on software quality. These results suggest that the positive effects of client coordination on software quality are offset when interorganizational differences diminish the communication quality between the vendor and the client. Interestingly, none of the global boundary variables had an interaction effect on software quality as we predicted in hypotheses 4 a and 5 and only temporal boundaries had a main effect.

With respect to the development speed model, the predictive power of the regression model also increased significantly ($p < 0.001$) when the interaction terms were added to the model. The results are also interesting because prior research on the effect of team size on speed has been mixed. Adding developers brings more productive resources into the task, but it also creates more dependency links and coordination overhead [68]. Our results show that team size has a direct positive impact on development speed, but its interaction with vendor-team coordination is negative, suggesting that increased team size makes vendor coordination less effective in reducing speed. We say this with caution because there was no direct effect of coordination on speed, but our results suggest that team coordination in large teams actually reduce speed, which provides partial support for Hypothesis 6. However, we did not find support for Hypothesis 4b about the interaction effects of vendor-team coordination and temporal boundaries on software development speed.

B. Discussion

The findings of this study offer fresh insights into the role of client and vendor-team coordination in the global delivery of software services. In particular, our results raise awareness of the fact that in outsourcing contexts it is not sufficient to study general coordination—we need to distinguish coordination effects between the various groups involved in the collaborative task. We find that the use of coordination mechanisms affect software quality and development speed differently. While conventional wisdom emphasizes the beneficial role of coordination in global work, our results indicate that project managers are faced with a tradeoff with respect to investments in coordination. Investments in client and vendor-team coordination lead to a more direct effect on quality. Therefore, in projects where quality is paramount, it is worthwhile for the manager to invest in coordination mechanisms. On the other hand, our analysis shows that

there do not appear to be any impacts of coordination mechanisms on development speed. In fact, the use of vendor-team coordination mechanisms tends to reduce development speed somewhat, but not significantly. This finding is consistent with prior research which found that geographically distributed collaboration is associated with larger teams sizes (e.g., due to the need of redundant roles and liaisons) and when team size is controlled for, the effects associated with geographic dispersion go away [13].

While seemingly counterintuitive, this result is not entirely surprising. Although we do not address this directly, there is an implied cost attached to vendor-team coordination that manifests itself as significant overhead. This overhead generally consumes effort and time without necessarily leading to more software developed. However, these efforts bear fruit in terms of higher quality, all else being equal. Therefore, managers need to take the differing effects of coordination on quality and development speed into account in establishing appropriate coordination mechanisms for projects. Also, software projects have different types of dependencies [31], so the effectiveness of coordination may vary depending upon which types of dependencies are more critical to performance. For example, technical dependencies among software parts need to be managed to ensure that they work well together and are, therefore, more closely associated with quality. Temporal dependencies among various activities, on the other hand, have to do with synchronization of schedules among various activities and are more closely associated with development speed. The fact that coordination affected quality but not speed suggests that coordination is more important in offshore software service delivery for the management of technical dependencies than for temporal dependencies, or perhaps that temporal dependencies are effectively managed by other means (e.g., project plans, configuration management systems).

Our study of global boundaries offers additional interesting insights. A key finding is the significant interaction between client coordination and client communication barriers on quality, suggesting that when interorganizational differences affect the quality of communication with the client they reduce the effectiveness of coordination with respect to quality, as shown in Fig. 2. Another interesting result is the direct effect of the differences in time zones on quality. Espinosa and Carmel [56] discuss the pathways through which time-zone differences can hinder information sharing, issue resolution and clarification processes. In an experimental study, Espinosa *et al.* [84] also found that quality deteriorates with time zones. For Indian vendors, clients located in Singapore and Japan provide some overlap in working hours but not so with clients in Europe or North America. Although the literature highlights the benefits of “24-hour workdays,” our analysis shows that these benefits involve trade-offs with respect to quality.

The direct effect of temporal boundaries on development speed also provides seemingly counterintuitive results. Our analysis shows that temporal boundaries actually increase the development speed of the team, in contrast to the negative effect of these boundaries on quality. However, this is consistent with “follow-the-sun” practices discussed by Carmel [11]. We speculate that the difference in time zones may allow managers to

leverage the “24 hour work-day” in making more effective use of calendar time on a project. These duration advantages are greater when the time separation is larger, i.e., India and the US versus India and Singapore. Another possible explanation for the gain in development speed may lie in reduction in interruptions from the client-side that allow vendor teams to work efficiently on larger subtasks.

VI. CONCLUSION

Our study contributes to both practice and research, and is among the first to analyze coordination in offshore software services delivery from the vendor’s perspective, taking into account both vendor team and client coordination. While our study has obvious implications for offshore software service vendors, it also has important implications for client organizations because it sensitizes them to the tradeoffs between quality and development speed. Our findings suggest that coordination mechanisms can be effective to attain high quality (i.e., managing technical dependencies) but may lead to overheads that do not allow for gains in speed (i.e., managing temporal dependencies). Therefore, for mission-critical applications where reliability and response times are vital, the stress should be on ensuring good coordination with the client and within the project team. For applications where speed and time to market are critical, effective mechanisms like configuration management systems, optimal team size, and project management tools may be employed.

Our study provides further insights from the interaction effect between global boundaries and coordination. Although managers cannot entirely surmount the challenges associated with bridging these boundaries, they can mitigate them by instituting effective coordination mechanisms based on planning, feedback, and integrative roles. These mechanisms are all the more important in the global context where the role of informal, communication-oriented coordination is muted and the advantages of collocation are absent. Also, conditions where interorganizational differences may create client communication issues will require managers to closely monitor quality issues, while larger team sizes and working with organizations of very different cultures will require careful management of schedules.

Our findings in this study are not without some limitations. First, our sample is based on a relatively small number of projects at a few companies. However, since less than 1% of the software firms in India account for 60% of the export revenues [85] and all firms in our sample were at least CMM level 3, our results are representative of the Indian offshore context, which is a most popular destination for software outsourcing. While the sample is also skewed toward clients in North America, this too is representative of the Indian offshore model. However, generalizing beyond this setting has to be done with caution. For instance, our results may not generalize to settings where: firms are not CMM-certified; business software is not the dominant form of outsourcing; or firms operate in “niche” or specialized domains. Many firms in Eastern Europe operate in such markets and our results cannot seamlessly be extended to these settings.

Second, we used subjective assessment of key variables in our study. However, the use of key informants that possess special and relevant qualifications such as status, experience, or specialized knowledge is well-established in past research [86]. Third, several other factors could influence project performance, such as complexity [80] human resources [33] and the use of tools [43]. Given our focus on coordination, we did not include these factors in our models. More research is needed to examine these factors in detail. Finally, we identified a small set of contingencies that influence the efficacy of coordination, given the limitations of sample size. Many other coordination barriers exist and are not captured here. We hope that our work will spark interest in examining these in detail.

APPENDIX

MEASUREMENT SCALES

External (Client) Coordination: To what extent were the following true during the project?

- 1) One or more members of the project team had been formally designated to facilitate coordination with the client—ClientCoord1.
- 2) Projects documents and memos were shared regularly with the client—ClientCoord2.
- 3) Project milestones and delivery schedules were shared regularly with the client—ClientCoord3.
- 4) Error tracking procedures were established with the client—ClientCoord4.

Internal (Vendor Team) Coordination: To what extent were the following true during the project?

- 1) Project estimates (such as project size, cost and schedule) were documented regularly for use in planning and tracking the project—VendCoord1.
- 2) All project members agreed to the commitment of their time as per the project plan—VendCoord2.
- 3) The project’s actual results (such as schedule and cost) were compared regularly with estimates in the project plan—VendCoord3.
- 4) Corrective action was taken proactively when actual results deviated from the project plan—VendCoord4.
- 5) The project followed documented processes for software development—VendCoord5.

Client Communication Barriers: To what extent were the following true during the project?

- 1) We faced problems in communicating with the client due to language differences—Boundary1.
- 2) We faced problems in communicating with the client due to differences in understanding business practices—Boundary2.
- 3) We faced problems in communicating with the client due to differences in design and development methods—Boundary3.

Software Quality: Extent to which the project met the following client requirements.

- 1) Response time.
- 2) Flexibility.
- 3) Usability.

4) Reliability.

[All items above used seven-point Likert scales ranging from 7 = "Strongly Agree" to 1 = "Strongly Disagree"].

REFERENCES

- [1] E. Schwartz. (2008, Aug. 25). IT outsourcing gone bad: 4 painful lessons. *CIO Magazine*. Available: http://www.pcworld.com/businesscenter/article/150266/painful_lessons_from_it_outsourcing_gone_bad.html.
- [2] P. M. Leonardi and D. E. Bailey, "Transformational technologies & the creation of New Work practices: Making implicit knowledge explicit in task-based offshoring," *MIS Q.*, vol. 32, pp. 411–436, 2008.
- [3] H. H. Olsson, E. O. Conchúir, P. J. Ågerfalk, and B. Fitzgerald, "Two-stage offshoring: An investigation of the Irish bridge," *MIS Q.*, vol. 32, pp. 257–279, 2008.
- [4] R. Sabherwal, "The evolution of coordination in outsourced software development projects: A comparison of client and vendor perspectives," *Inf. Org.*, vol. 13, pp. 153–202, 2003.
- [5] I. Oshri, J. Kotlarsky, and L. P. Willcocks, "Managing dispersed expertise in IT offshore outsourcing: Lessons from tata consultancy services," *MISQ Exe.*, vol. 6, pp. 53–65, 2007.
- [6] W. Orlikowski, "Knowing in practice: Enacting a collective capability in distributed organizing," *Org. Sci.*, vol. 13, pp. 249–273, May/Jun. 2002.
- [7] M. B. Watson-Manheim, K. M. Chudoba, and K. Crowston, "Discontinuities and continuities: A new way to understand virtual work," *Inf. Technol. People*, vol. 15, pp. 191–209, 2002.
- [8] S. Gopalakrishnan, V. P. Kochikar, S. Yegneshwar, "The offshore model for software development: The Infosys experience," presented at the *ACM SIGCPR/SIGMIS Conf. Comput. Personnel Res.*, Denver, CO, 1996.
- [9] A. Gopal, T. Mukhopadhyay, and M. S. Krishnan, "The role of software processes & communication in offshore software development," *Commun. ACM*, vol. 45, pp. 193–200, 2002.
- [10] V. Couto, M. Mani, A. Lewin, and C. Peeters, "The globalization of white-collar work," Offshoring Res. Netw. Working Paper, CIBER, Duke University, Durham, NC, 2006.
- [11] E. Carmel, *Global Software Teams*. Upper Saddle River, NJ: Prentice Hall, 1999.
- [12] J. D. Herbsleb and R. E. Grinter, "Architectures, coordination, and distance: Conway's law and beyond," *IEEE Softw.*, vol. 16, no. 5, pp. 63–70, Sep./Oct. 1999.
- [13] J. D. Herbsleb and A. Mockus, "An empirical study of speed and communication in globally distributed software development," *IEEE Trans. Softw. Eng.*, vol. 29, pp. 481–494, Jun. 2003.
- [14] M. C. Lacity and R. Hirschheim, "The information-systems outsourcing bandwagon," *Sloan Manage. Rev.*, vol. 35, no. 1, pp. 73–86, 1993.
- [15] J. Kotlarsky, P. C. van Fenema, and L. P. Willcocks, "Developing a knowledge-based perspective on coordination: The case of global software projects," *Inf. Manage.*, vol. 45, pp. 96–108, 2008.
- [16] T. Malone and K. Crowston, "The interdisciplinary study of coordination," *ACM Comput. Surveys*, vol. 26, pp. 87–119, 1994.
- [17] A. Bednars, "The downside of offshoring," *Netw. World*, vol. 33, pp. 33–35, 2004.
- [18] S. Thurm, "Tough shift—Lesson in India for Valicert," *Wall Street J. New York*, vol. A1, p. 1, 2004.
- [19] R. Sabherwal and V. Choudhury, "Governance of remotely outsourced software development: A Comparison of client and vendor perspectives," in *Information Systems Outsourcing*, R. Hirschheim, A. Heinzl, and J. Dibbern, Eds. Heidelberg, Germany: Springer, 2006, pp. 187–222.
- [20] R. Kraut and L. Streeter, "Coordination in software development," *Commun. ACM*, vol. 38, pp. 69–81, 1995.
- [21] A. H. Van de Ven, L. A. Delbecq, and R. J. Koenig, "Determinants of coordination modes within organizations," *Am. Sociol. Rev.*, vol. 41, pp. 322–338, Apr. 1976.
- [22] N. Levina and J. W. Ross, "From the vendor's perspective: Exploring the value proposition in information technology outsourcing," *MIS Q.*, vol. 27, pp. 331–364, 2003.
- [23] D. B. Klein, "Convention, social order, and the two coordinations," *Constitutional Pol. Econ.*, vol. 8, pp. 319–335, 1997.
- [24] D. Lau and J. K. Murnighan, "Demographic diversity and faultlines: The compositional dynamics of organizational groups," *Acad. Manage. Rev.*, vol. 23, pp. 325–340, 1998.
- [25] J. March and H. A. Simon, *Organizations*. New York: John Wiley and Sons, 1958.
- [26] V. Choudhury and R. Sabherwal, "Portfolios of control in outsourced software development projects," *Inf. Syst. Res.*, vol. 14, pp. 291–314, Sep. 2003.
- [27] K. Crowston and E. E. Kammerer, "Coordination and collective mind in software requirements development," *IBM Syst. J.*, vol. 37, pp. 227–245, 1998.
- [28] S. R. Nidumolu, "The effect of coordination and uncertainty on software project performance: Residual performance risk as an intervening variable," *Inf. Syst. Res.*, vol. 6, pp. 191–219, 1995.
- [29] M. Hoegl and H. G. Gemuenden, "Teamwork quality and the success of innovative projects: A theoretical concept and empirical evidence," *Org. Sci.*, vol. 12, pp. 435–449, Jul. 2001.
- [30] J. Cummings, A. Espinosa, and C. Pickering, "Crossing spatial and temporal boundaries in globally distributed projects: A relational model of coordination delay," *Inf. Syst. Res.*, vol. 20, pp. 420–439, Sep. 2009.
- [31] J. A. Espinosa, S. A. Slaughter, R. E. Kraut, and J. D. Herbsleb, "Team knowledge and coordination in geographically distributed software development," *J. Manage. Inf. Syst.*, vol. 24, pp. 135–169, Sum. 2007.
- [32] J. G. Coopridge and J. C. Henderson, "Technology-process fit: Perspectives on achieving prototyping effectiveness," *J. Manage. Inf. Syst.*, vol. 7, pp. 67–87, 1991.
- [33] R. Pressman, *Software Engineering: A Practitioner's Approach*. New York: McGraw Hill, 2001.
- [34] P. Thibodeau and L. Rosencrance, "Users losing billions due to bugs," *Computerworld*, vol. 36, pp. 1–2, 2002.
- [35] S. K. Ethiraj, P. Kale, M. S. Krishnan, and J. V. Singh, "Where do capabilities come from and how do they matter? A study in the software services industry," *Strategic Manage. J.*, vol. 26, pp. 25–45, 2005.
- [36] L. Wallace, M. Keil, and A. Rai, "How software project risk affects project performance: An investigation of the dimensions of risk and an exploratory model," *Decis. Sci.*, vol. 35, pp. 289–321, 2004.
- [37] T. Ravichandran and A. Rai, "Quality management in systems development: An organizational system perspective," *MIS Q.*, vol. 24, pp. 381–415, 2000.
- [38] A. Tiwana, "An empirical study of the effect of knowledge integration on software development performance," *Inf. Softw. Technol.*, vol. 46, pp. 899–906, 2004.
- [39] J. A. Espinosa, S. A. Slaughter, R. E. Kraut, and J. D. Herbsleb, "Familiarity, complexity and team performance in geographically distributed software development," *Org. Sci.*, vol. 18, pp. 613–630, Jul./Aug. 2007.
- [40] B. Curtis, H. Krasner, and N. Iscoe, "A field study of the software design process for large systems," *Commun. ACM*, vol. 31, pp. 1268–1287, 1988.
- [41] J. Kotlarsky and I. Oshri, "Social ties, knowledge sharing & successful collaboration in globally distributed system development projects," *Eur. J. Inf. Syst.*, vol. 14, pp. 37–48, 2005.
- [42] J. D. Herbsleb, A. Mockus, T. A. Finholt, and R. E. Grinter, "An empirical study of global software development: Distance and speed," presented at the Int. Conf. Softw. Eng., Toronto, ON, Canada, 2001.
- [43] M. S. Krishnan, C. H. Kriebel, S. Kekre, and T. Mukhopadhyay, "An empirical analysis of productivity and quality in software products," *Manage. Sci.*, vol. 46, pp. 745–759, 2000.
- [44] P. Vlaar, P. C. van Fenema, and V. Tiwari, "Co-creating understanding and value in distributed work: How members of onsite and offshore vendor teams give, make, denab and break sense," *MIS Q.*, vol. 32, 2008.
- [45] C. H. Loch, C. Terwiesch, and S. Thomke, "Parallel and sequential testing of design alternatives," *Manage. Sci.*, vol. 47, pp. 663–678, May 2001.
- [46] H. Mintzberg, *Mintzberg on Management*. New York: The Free Press, 1989.
- [47] D. G. Ancona and D. F. Caldwell, "Bridging the boundary: External activity and performance in organizational teams," *Admin. Sci. Q.*, vol. 37, pp. 634–665, 1992.
- [48] R. E. Kraut and L. A. Streeter, "Coordination in software development," *Commun. ACM*, vol. 38, pp. 69–81, 1995.
- [49] V. S. Mookerjee and I. R. Chiang, "A dynamic coordination policy for software system construction," *IEEE Trans. Softw. Eng.*, vol. 28, pp. 684–694, Jul. 2002.
- [50] A. Gopal and S. Gosain, "Research note—the role of organizational controls and boundary spanning in software development outsourcing: Implications for project performance," *Inf. Syst. Res.*, vol. 21, no. 4, pp. 960–982, Dec. 2010.
- [51] R. M. Krauss and S. R. Fussell, "Mutual knowledge and communicative effectiveness," in *Intellectual Teamwork*, G. Jolene, *et al.*, Ed. Hillsdale, NJ: L. Erlbaum Associates, Inc., 1990, pp. 111–145.

- [52] S. Sarker and S. Sahay, "Implications for space & time for distributed work: An interpretive study of US-Norwegian systems development teams," *Eur. J. Inf. Syst.*, vol. 13, pp. 3–20, 2004.
- [53] S. Krishna, S. Sahay, and G. Walsham, "Managing cross-cultural issues in global software outsourcing," *Commun. ACM*, vol. 47, pp. 62–66, 2004.
- [54] S. Sahay, B. Nicholson, and S. Krishna, *Global IT Outsourcing*. Cambridge, U.K.: Cambridge Univ. Press, 2003.
- [55] M. L. Maznevski and K. M. Chudoba, "Bridging space over time: Global virtual team dynamics and effectiveness," *Org. Sci.*, vol. 11, pp. 473–492, 2000.
- [56] J. A. Espinosa and E. Carmel, "The impact of time separation on coordination in global software teams: A conceptual foundation," *J. Softw. Process: Practice Improvement*, vol. 8, pp. 249–266, Sep. 2004.
- [57] P. C. van Fenema and S. Qureshi, "A phenomenological exploration of adaptation in a polycontextual work environment," presented at the 37th Hawaiian Int. Conf. Syst. Sci., Big Island, HI, 2004.
- [58] K. Knoll and S. L. Jarvenpaa, "Working together in global virtual teams," in *The Virtual Workplace*. Hershey, PA: IDEA Group, 1998, pp. 2–23.
- [59] S. Moore, *Cultural Challenges In Offshore Outsourcing*: Forrester Research, 2004.
- [60] G. Hofstede, *Cultures & Organisations*, U.K.: McGraw-Hill, 1991.
- [61] G. Walsham, "Cross-cultural software production and use: A structural analysis," *MIS Q.*, vol. 26, pp. 359–380, 2002.
- [62] P. Morosini, S. Shane, and H. Singh, "National cultural distance and cross-border acquisition performance," *J. Int. Bus. Stud.*, vol. 29, pp. 137–158, 1998.
- [63] A. Roland, *In Search of Self in India & Japan*. Princeton, NJ: Princeton Univ. Press, 1998.
- [64] M. Kobayashi-Hillary, "A passage to India," *ACM Queue*, pp. 55–60, 2005.
- [65] M. Barrett, A. Drummond, and S. Sahay, "Exploring the impact of cross-cultural differences in international software development teams: Indian expatriates in Jamaica," presented at the Fourth Eur. Conf. Inf. Syst., Lisbon, Portugal, 1996.
- [66] M. Keil, B. C.Y. Tan, K. K. Wei, and T. Saarinen, "A cross-cultural study on escalation of commitment behavior in software projects," *MIS Q.*, vol. 24, pp. 299–325, 2000.
- [67] S. Overby, "The hidden costs of offshore outsourcing," *CIO Mag.*, Sep. 1, 2003, pp. 60–66.
- [68] F. Brooks, in *The Mythical Man-Month: Essays on Software Engineering*, Anniversary, Ed. Reading, MA: A. Wesley, 1995.
- [69] S. Faraj and L. Sproull, "Coordinating expertise in software development teams," *Manage. Sci.*, vol. 46, pp. 1554–1568, Dec. 2000.
- [70] G. B. Davis, P. Ein-Dor, W.R. King, and R. Torkzadeh, "IT offshoring: History, prospects and challenges," *J. Assoc. Inf. Syst.*, vol. 7, pp. 770–795, 2006.
- [71] S. Szejko, "An exercise in evaluating significance of software quality criteria," *ACM SIGCSE Bull.*, vol. 31, p. 199, 1999.
- [72] A. Gopal, T. Mukhopadhyay, M. S. Krishnan, and D.R. Goldenson, "Measurement programs in software development: Determinants of success," *IEEE Trans. Softw. Eng.*, vol. 28, no. 9, pp. 863–875, Sep. 2002.
- [73] M. Hayes, "Precious Connection," *Inf. Week.*, pp. 34–50, 2003.
- [74] P. Thibodeau, "More IT jobs go offshore," *Computerworld*, vol. 38, p. 13, 2004.
- [75] B. Kogut and H. Singh, "The effect of national culture on the choice of entry mode," *J. Int. Bus. Stud.*, vol. 19, pp. 411–432, 1988.
- [76] J. F. Hair, R. E. Anderson, R. L. Tatham, and W. C. Black, *Multivariate Data Analysis*. Englewood Cliffs, NJ: Prentice-Hall, 1995.
- [77] N. Venkatraman, "Strategic orientation of business enterprises: The construct, dimensionality, and measurement," *Manage. Sci.*, vol. 35, pp. 942–962, 1989.
- [78] L. Hu and P. M. Bentler, *Evaluating model fit. Structural Equation Modeling: Concepts, Issues & Applications*. Thousand Oaks, CA: SAGE Publications, 1995.
- [79] P. M. Podsakoff and D. W. Organ, "Self-reports in organizational research: Problems and prospects," *J. Manage.*, vol. 12, pp. 531–544, Winter 1986.
- [80] B. R. Boehm, *Software Engineering Economics*. Englewood Cliffs, N.J.: Prentice-Hall, 1981.
- [81] D. A. Belsley, E. Kuh, and R. E. Welsch, *Regression Diagnostics: Identifying Influential Data and Sources of Collinearity*. New York: John Wiley & Sons, 1980.
- [82] L. Aiken and S. West, *Multiple Regression: Testing and Interpreting Interactions*. Newbury Park, CA: Sage, 1991.
- [83] T. A. Carte and C. J. Russell, "In pursuit of moderation: Nine common errors and their solutions," *MIS Q.*, vol. 27, pp. 479–501, 2003.

- [84] J. A. Espinosa, N. Nan, and E. Carmel, "Do gradations of time separation make a difference in performance? A first laboratory study," in *Proc. 2nd IEEE Int. Conf. Global Softw. Eng.*, Munich, Germany, 2007, pp. 12–20.
- [85] NASSCOM, "Indian IT Industry: A Success Story," 2004.
- [86] A. H. Segars and V. Grover, "Strategic information systems planning success: An investigation of the construct and its measurement," *MIS Q.*, vol. 22, pp. 139–163, 1998.



engineering.

Anandasivam Gopal received the Ph.D. degree in information systems from the Tepper School of Business, Carnegie Mellon University, Pittsburgh, PA.

He is currently an Associate Professor at the Robert H. Smith School of Business, University of Maryland, College Park. He is an Associate Editor of *Information Systems Research* and a Guest Associate Editor of *Management Science*. His current research interests include the economics of IS, focusing on software-development outsourcing and offshoring, venture capital, and empirical software



laboration tasks like enterprise architecture.

J. Alberto Espinosa received the Ph.D. degree in information systems from the Tepper School of Business, Carnegie Mellon University, Pittsburgh, PA.

He is currently an Associate Professor of information technology at the Kogod School of Business, American University, Washington, DC. His current research interests include coordination and performance in global technical projects across global boundaries, particularly spatial and temporal distance, and coordination of technical work across time zones and coordination in large-scale technical col-



design, use, and strategic leverage in corporate settings, and also in examining social and strategic issues related to organizational transformations as well as coordination issues in software development.

Sanjay Gosain received the Ph.D. degree in information systems from the Marshall School of Business, University of Southern California.

He is currently in a web strategy and planning role with the Capital Group Companies, Inc., Irvine, CA, a privately held investment organization. He was an Assistant Professor of information systems in the Decision and Information Technologies Department, Robert H. Smith School of Business, University of Maryland. His current research interests include the drivers of effective information technology (IT) design, use, and strategic leverage in corporate settings, and also in examining social and strategic issues related to organizational transformations as well as coordination issues in software development.



David P. Darcy received the Ph.D. degree from the Joseph M. Katz Graduate School of Business, University of Pittsburgh, Pittsburgh, PA.

He is currently a Program Director at the Irish Management Institute, Dublin, Ireland. He was an Assistant Professor of information systems in the Decision and Information Technologies Department, Robert H. Smith School of Business, University of Maryland. His current research interests include software development and software complexity, and open source software and programmer learning.